

CRYSS Input Parser Pipeline

Technical Overview

CRYSS Project

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1 Processing Pipeline Overview

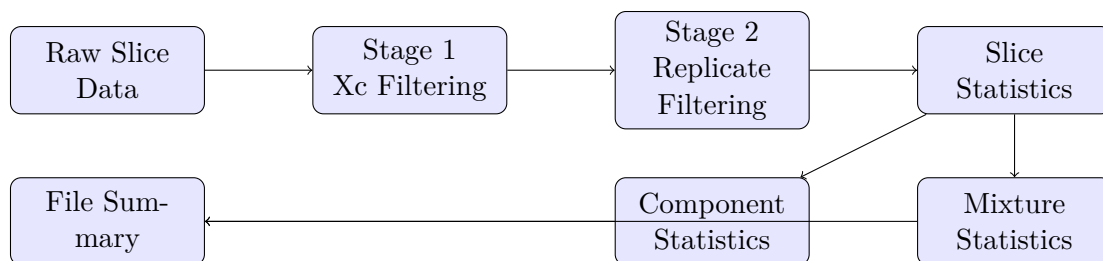


Figure 1: CRYSS Input Parser Processing Pipeline

2 Glossary of Terms

Xs Solid-phase composition. A vector of mole fractions satisfying $\sum_i X_s^{(i)} = 1$.

Xl Liquor-phase composition. A vector of mole fractions satisfying $\sum_i X_l^{(i)} = 1$.

Xc Mixture composition. The feed composition for the slice. Satisfies $\sum_i X_c^{(i)} = 1$.

R Solid fraction. The mass or mole fraction of solid in the slice.

Slice A single experimental measurement consisting of (X_s, X_l, X_c, R) .

Mixture A group of slices sharing the same mixture identifier.

Replicate A slice that repeats the same mixture and slice number (e.g. EXP001F5).

Canonical Values Median values across accepted slices.

3 Replicate Filtering Logic

Replicate filtering ensures that only chemically consistent slices contribute to mixture-level statistics. The process is defined as follows.

3.1 Canonical Values

Canonical mixture values are computed using medians across slices:

$$X_{c,\text{canonical}}^{(i)} = \text{median}(X_c^{(i)}), \quad R_{\text{canonical}} = \text{median}(R).$$

3.2 Acceptance Criteria

A slice is accepted if:

$$\max_i \left| X_{c,\text{slice}}^{(i)} - X_{c,\text{canonical}}^{(i)} \right| < 0.02,$$

$$|R_{\text{slice}} - R_{\text{canonical}}| < 0.10.$$

Slices failing either condition are rejected.

3.3 Rationale

The X_c tolerance removes analytical noise in mixture composition. The R tolerance removes noise in solid fraction determination. Together, they ensure robust mixture-level aggregation.

4 Slice Geometry Metrics

Slice geometry quantifies the separation strength and purity of each slice.

4.1 Solid Enrichment Index

$$SE = \sum_i (X_s^{(i)} - X_c^{(i)})^2.$$

Measures enrichment of components in the solid phase.

4.2 Liquor Enrichment Index

$$LE = \sum_i (X_l^{(i)} - X_c^{(i)})^2.$$

Measures depletion of components in the liquor phase.

4.3 Solid–Liquor Separation

$$SLS = \sum_i (X_s^{(i)} - X_l^{(i)})^2.$$

Measures the sharpness of the slice.

4.4 Slice Purity Index

$$PI = \max_i X_s^{(i)}.$$

Measures dominance of the major solid-phase component.

5 Excel Workbook Structure

The parser produces a multi-sheet Excel workbook summarising all intermediate and final statistics.

1. **slice_summary**: canonical slice values
2. **mixture_stats**: mixture-level medians, variation, purity
3. **slice_stats**: slice-level geometry metrics
4. **component_stats**: per-component trends
5. **file_summary**: global metadata and QC totals

This ordering mirrors the processing pipeline:

$$\text{slice} \rightarrow \text{mixture} \rightarrow \text{component} \rightarrow \text{file}.$$

6 Worked Example

Consider mixture 260829_002, containing four slices. After Xc filtering, all slices are accepted. After replicate filtering, one slice is rejected due to excessive deviation in R .

Canonical mixture values are computed from the remaining slices:

$$R_{\text{mix}} = 0.7940, \quad X_{s,\text{mix}} = \text{median}(X_s), \quad X_{l,\text{mix}} = \text{median}(X_l).$$

Slice-to-slice variation is quantified using standard deviations. Lever-rule consistency is evaluated using:

$$X_{c,\text{mix}} = X_{s,\text{mix}} + X_{l,\text{mix}}.$$

The mixture purity index is:

$$PI_{\text{mix}} = \text{mean}(|X_s - X_l|).$$

This example illustrates the full QC and aggregation pipeline.